

Decentralized Proof of Personhood

Audio Liveness Challenge Module — Work Completed / Progress & References

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Work Completed / Progress

Modules Completed

#	Module	Stack	Status
1	Smart Contract (Soulbound Token)	Solidity, Hardhat, OpenZeppelin (ERC-721 + EIP-5192)	Complete.
2	Backend Verification API	Python, FastAPI, web3.py	Complete.
3	IPFS Metadata Pinning	Pinata API	Complete
4	Frontend — Wallet Connect	React, Vite, ethers.js	Complete.
5	Frontend — Visual Liveness Challenge	MediaPipe Face Landmarker (on-device, WASM/GPU)	Complete
6	Frontend — Result / Badge Screen	React	Complete.
7	Audio Liveness Challenge (random sentence read-aloud)	Web Audio API / MediaRecorder + ASR	Not yet implemented — design stage.

Current Implementation Status

The visual (gesture-based) liveness pipeline and the full mint pipeline (challenge → verify → IPFS → Soulbound Token) are functional end-to-end and demo-able today. The system currently supports one liveness modality (vision); this phase adds a second, independent modality (voice), following the same "nothing leaves the device unverified" design principle already used for the camera flow.

Planned implementation, which slots directly into the existing architecture:

- **Backend:** extend the `CHALLENGES` list with an `audio` type carrying a randomly selected sentence; add a verification step that runs speech-to-text on the submitted clip and scores it against the prompted sentence (word-error-rate / edit-distance threshold) plus basic replay/spoof heuristics (spectral flatness, compression-artifact and energy-variance checks) before authorizing a mint — mirroring how the `verify` endpoint already gates on the visual challenge.

- **Frontend:** a new `AudioChallengeRecorder.jsx` component (parallel to the existing `ChallengeRecorder.jsx`) to capture the mic clip, show a waveform/recording indicator, and submit it alongside the wallet address.
- **Privacy stance:** transcription/scoring is intended to run within the request cycle and discard the raw audio immediately after scoring (not persisted), consistent with the project's no-raw-biometric-storage claim.

Current status: architecture and approach finalized; implementation in progress, targeting the same two integration points (`/challenge`, `/verify`) used by the visual flow.

References

Core anti-spoofing / voice liveness literature

- [1] N. Evans, T. Kinnunen, and J. Yamagishi, "Spoofing and countermeasures for automatic speaker verification," in *Proc. Interspeech*, 2013, pp. 925–929.
- [2] Z. Wu, N. Evans, T. Kinnunen, J. Yamagishi, F. Alegre, and H. Li, "Spoofing and countermeasures for speaker verification: A survey," *Speech Communication*, vol. 66, pp. 130–153, 2015.
- [3] Z. Wu *et al.*, "ASVspoof: The automatic speaker verification spoofing and countermeasures challenge," *IEEE J. Sel. Topics Signal Process.*, vol. 11, no. 4, pp. 588–604, Jun. 2017.
- [4] T. Kinnunen *et al.*, "The ASVspoof 2017 challenge: Assessing the limits of replay spoofing attack detection," in *Proc. Interspeech*, 2017, pp. 2–6.
- [5] X. Wang, J. Yamagishi, *et al.*, "ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech," *Computer Speech & Language*, vol. 64, 2020, Art. no. 101114.
- [6] M. Todisco, H. Delgado, and N. Evans, "Constant Q cepstral coefficients: A spoofing countermeasure for automatic speaker verification," *Computer Speech & Language*, vol. 45, pp. 516–535, 2017.

Text-dependent / text-prompted verification (random sentence read-aloud)

- [7] A. Larcher, K. A. Lee, B. Ma, and H. Li, "Text-dependent speaker verification: Classifiers, databases and RSR2015," *Speech Communication*, vol. 60, pp. 56–77, 2014.
- [8] K. A. Lee *et al.*, "The RedDots data collection for speaker recognition," in *Proc. Interspeech*, 2015, pp. 2996–3000.

Datasets

- [9] A. Nagrani, J. S. Chung, and A. Zisserman, “VoxCeleb: A large-scale speaker identification dataset,” in *Proc. Interspeech*, 2017, pp. 2616–2620.
- [10] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, “Librispeech: An ASR corpus based on public domain audio books,” in *Proc. IEEE ICASSP*, 2015, pp. 5206–5210.

Speech-to-text engine (for the sentence-match check itself)

- [11] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” *arXiv preprint arXiv:2212.04356*, 2022.

Relevant conferences/venues (where references [1]–[8] above were published, useful for a "field of application" line in the report): Interspeech, IEEE ICASSP, The Speaker and Language Recognition Workshop (Odyssey), and the IEEE International Joint Conference on Biometrics (IJCB).